



Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

*Department of Mechanical and Manufacturing
Engineering Trinity College Dublin,
The University of Dublin*

ME 4B7 Computer Aided Design

CAD Assignment 1: Submersible Design

Robert Dillon 22336720
Magnus Runham 22336288
Finn O'Connor 22336740
Micheal Sadlier 22336873
Peter Hamilton 22337384

13/10/2025

Table of Contents

1. OBJECTIVES 3

2. FINAL DESIGNERROR! BOOKMARK NOT DEFINED.

 2.1 WINDOW GEOMETRY 4

 2.2 CHAMBER GEOMETRY 4

3. MATERIAL ANALYSIS 4

 3.1 HULL MATERIAL..... 4

 3.1.1 Proposed Materials 5

 3.1.2 Selected Material 5

 3.1.3 Fatigue and Corrosion 5

 3.1.4 Lifecycle 6

 3.2 VIEWPORT MATERIAL 6

 3.3 O-RING MATERIAL 7

 3.4 BUOYANCY 7

4. STRUCTURAL ANALYSIS..... 8

 4.1 MESH REFINEMENT 8

 4.2 STATIC ANALYSIS 8

 4.3 BUCKLING ANALYSIS 10

5. O-RING DESIGN 10

 5.1 MODEL SETUP 11

 5.2 RESULTS 12

REFERENCES..... 12

APPENDIX A: STRUCTURAL ANALYSIS 13

APPENDIX B: O-RING DESIGN 15

1. Objectives

The goal of this project is to design a submersible pressure vessel with a capacity of two occupants. Our design will travel to a depth of 300m and operate safely at a hydrostatic pressure of 3.02 MPa. The spherical chamber of the submersible will have a 1.8m internal diameter. The design conditions are as follows:

- Design depth = 300m
- Design pressure = 3.02MPa
- Number of occupants = 2
- Vessel diameter = 1.8m
- Ambient temperature of water = 4-15°C
- Cabin internal temperature = 25°C

Design approach:

- Determine design conditions
- Determine vessel geometry aligning with PHVO-1 safety standards
- Review suitable materials for window and hull
- Run simulations on SolidWorks to examine our chosen window and hull design
- Determine O-ring size shape and groove
- Review suitable materials for the O-ring
- Run simulations on SolidWorks to examine our chosen O-ring design

2. Final Design

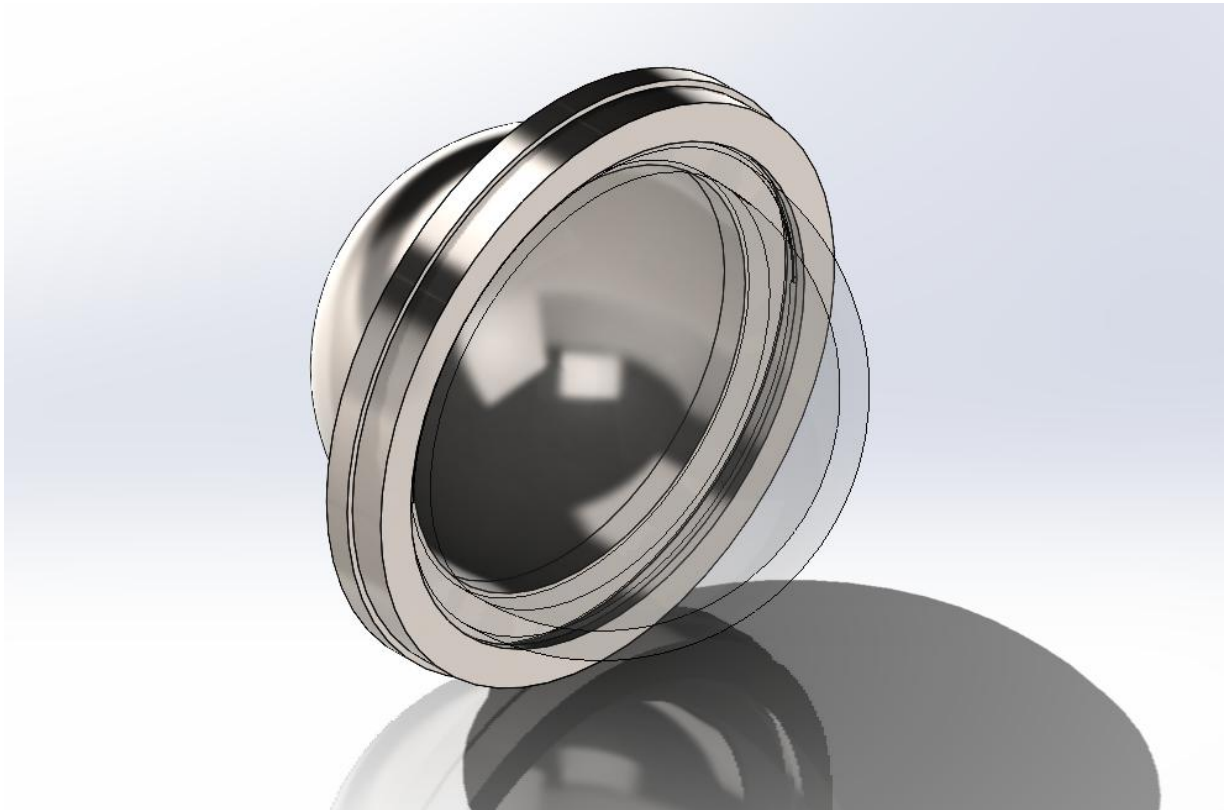


Figure 1: Proposed design of submersible compression chamber and window

2.1 Window Geometry

The design conditions for the window are as follows:

- Design pressure (P) = 3.02MPa
- Design temperature = 20°C - 25°C, accounting for the internal temperature of the chamber
- Window internal diameter (D_i) = 1.8m, large enough for two people

The hemi-spherical window with an equatorial flange was selected from the list of standard geometries outlined in PVHO-1 – Fig. 2-2.2.1-3 (a) standard window geometries. From PVHO-1 – Table 2-2.3.1-4 a conversion factor (CF) = 9 and $N = 1$ were selected for the calculation of the short-term critical pressure (SCTP).

$$STCP = CF \times P = 9 \times 3.02 = 27\text{MPa}$$

Calculating the dimensionless ratio for windows:

$$T/D_i = 0.06 \text{ for a STCP of 27MPa with an angle of 180 degrees, from PVHO-1 – Fig. 2-2.5.1-7}$$

From here the minimum acceptable thickness of the window (t) can be calculated:

$$t = 0.06(D_i) = 10.8\text{cm}$$

Based on this calculation complying with PVHO-1 standards, a window thickness of 11cm was selected for our design.

2.2 Chamber Geometry

From our studies in 3B3 Mechanics of Solids we learned about the dissipation of stress on pressurised vessels. In a spherical vessel the membrane stress is equal in every direction, with equal principal stress and no in-plane stress ($\sigma = \frac{pr}{2t}$ all around), whereas a thin cylindrical vessel would have a larger hoop stress and a smaller longitudinal stress ($\sigma_1 = \frac{pr}{t}$ and $\sigma_2 = \frac{pr}{2t}$). Because the sphere essentially avoids this split by keeping all sections as principal planes, it distributes stress more uniformly than a cylindrical vessel. This leads to a reduction in stress on wall stress for the same equivalent pressure as its cylindrical counterpart allowing for a reduction in the amount of material required to maintain a constant internal pressure. As such, the decision was made that our submersible would be spherical in nature. Through a series of trial and error a chamber wall thickness of 5cm was selected for our design. As outlined in PVHO-1 Fig. 2-2.10.1-4, the internal diameter of the hull, $D_f = 1.02 \times D_i = 1.76\text{m}$. Through trial and error using a design study where the thickness was set as a global parameter, 5cm was determined to be a suitable thickness for the hull.

3. Material Analysis

3.1 Hull Material

Selecting the materials to be used in the hull (main body) for the submersible is critical to both the safety of the passengers and the performance of the vessel. It must resist external pressure at a depth of 300 m, which equates to about 3.02 MPa, while the density of the materials influences the vessel's buoyancy and influences its overall size and mass. Similarly, the materials must also have high fatigue resistance and high fracture toughness's to remain intact during repeated pressurisation cycles in corrosive sea water.

While ISO standards [1] recommend a hydrostatic test factor between 1.1 and 1.25 times the working pressure, a more conservative approach was chosen that follows the recommendations of an article from the Journal of Ocean Engineering and Marine Engineering [2], it specifies that ASME PVHO-1 classification standards require a collapse safety factor of around 1.5 for manned steel spherical pressure vessels. As a result, the design will use a test pressure of 3MPa and a minimum collapse pressure of 4.5MPa to provide additional assurance of structural safety.

3.1.1 Proposed Materials

The materials considered for the submersible design were HY-130 steel, 7075-T6 aluminium, carbon/epoxy composites and Ti-6Al-4V ELI titanium. Each offers a trade-off between density, elastic moduli, compressive strengths and corrosion resistance.

Table 1: Characteristics of potential materials

Material	Density (kg/m ³)	Young's Modulus (GPa)	Compressive Yield Strength (MPa) **	Corrosion Resistance
Steel HY-130	7885	207	920	Strong but prone to electrochemical corrosion
Aluminium 7075-T6	2810	71.7	503	Poor
Carbon/ Epoxy Composite	1600	70	570	Mediocre
Titanium 6Al-4V ELI	4430	113.8	810	Excellent

** Tensile strength figures were used as compressive figures varied; however, many sources cite that there is a negligible difference between tensile and compressive strength figures for these materials

- **Steel** provides excellent strength and stiffness but suffers from electrochemical corrosion in seawater.
- **7075-T6 Aluminium** is light but its inability to resist corrosion in chloride environments makes it a very poor choice in submersible applications.
- **Carbon/epoxy composites** are strong and lightweight but prone to moisture absorption, delamination, and degradation under prolonged hydrostatic cycling. When coupled with their relatively low compressive yield strength, these factors limit their suitability for deep submersible hulls.
- **Ti-6Al-4V ELI Titanium**, though costlier, combine low density, high strength and exceptional seawater corrosion resistance, which makes them ideal for long-term submersible structures.

References from above in no particular order (Zotero library corrupted): [3], [2], [4], [5], [6], [7], [8].

3.1.2 Selected Material

At 300 m depth, the external pressure (~3.02 MPa) demands materials with high compressive yield strength. The combination of a high elastic modulus and compressive strength makes Titanium 6Al-4V ELI favourable. This is due to the fact the fact that it can resist deformation under pressure while simultaneously providing early evidence of potential deformation. Similarly, its density (4,430 kg/m³) is significantly lower than the traditional most used material, HY-130 steel (7,885 kg/m) making it possible to reduce the overall weight of the submersible while ensuring buoyancy efficiency. The primary drawbacks for Aluminium 7075-T6 and carbon/epoxy is their respective lack of ability to resist corrosion in marine environments composites as they lack the long-term reliability needed under cyclic compressive hydrostatic loading.

3.1.3 Fatigue and Corrosion

Steel is strong but prone to electrochemical corrosion in seawater, requiring protective coatings. Aluminium 7075-T6 corrodes rapidly in real seawater environments, as a combination of saltwater and microorganisms in the sea increase surface activity and break down protective oxide films which leads to deeper localised corrosion. Carbon/epoxy composites take in moisture, leading to gradual weakening and loss of stiffness. Titanium 6Al-4V ELI, however, forms a stable oxide film that resists seawater corrosion and maintains performance after repeated cycles.

3.1.4 Lifecycle

Despite the high initial cost, titanium's combination of strength retention, negligible corrosion loss and minimal maintenance contributes to it being the most appropriate choice. To reinforce this point, steel and aluminium require continual inspection and protection renewal, while carbon /epoxy face degradation and uncertain reparability.

3.2 Viewport Material

The viewport material of a submersible matters for many different reasons. It needs to have very good structural integrity to withstand the immense hydrostatic pressure being exerted on it. It acts as a pressure boundary between the external hydrostatic pressure and the internal cabin pressure. If failure were to occur in the material, sudden catastrophic disaster as seen before in cases such as the OceanGate. It also needs to offer good optical transparency so that the passengers can manoeuvre it and observe their surroundings.

To assess how suitable the acrylic is, it must conform to the following performance requirements:

- **Pressure Resistance:** The viewport must safely withstand the hydrostatic pressure being exerted at 300m deep in the sea. The corresponding hydrostatic pressure at this operating depth is approximately 3.02 MPa. This also needs to be in accordance with safety standards such as ASME PVHO-1 (2007) and ISO 21173 to account for an appropriate factor of safety.
- **Optical Performance:** The material must also have good visible light transmittance with minimal distortion to ensure clear visibility and perception through the dome.
- **Creep Resistance:** The material must also not deform high hydrostatic cyclical loading over time. This may lead to the seals or structural integrity of the viewport to be compromised.
- **Thermal and Environmental Resistance:** The viewport material must withstand the operating temperatures of 4-15°C.
- **Manufacturability:** The material must also be manufactured and machined into a hemispherical geometry with uniform thickness and no defects.

After analysing and researching different materials that could be used for this application, the chosen material was “Transpalite SS Acrylic”, a cast polymethyl methacrylate (PMMA) derivative specifically engineered for high optical quality and mechanical reliability in marine and aerospace applications. It is often the material of choice for deep-sea subs due to its excellent light transmission, low density, chemical resistance, and ability to withstand massive amounts of cyclical hydrostatic loading.

The following table of properties was acquired through Stanley Plastics, a leading high-performance plastics manufacturer. The given values are based on tests done by the company during the R&D stages of “Transpalite SS” Acrylic [9].

Table 2: Material properties of Transpalite SS acrylic

Physical Property	ASME-PVHO-1-2012 Specified Values	Transpalite SS Example Values
Tensile		
a) Ultimate strength	≥ 62 MPa	72.9 MPa
b) Elongation at break	$\geq 2\%$	4.6%
c) Modulus of elasticity	≥ 2760 MPa	3338 MPa
Compressive		
a) Yield strength	≥ 103 MPa	110 MPa
b) Modulus of elasticity	≥ 2760 MPa	2760 MPa
Flexural ultimate strength	≥ 97 MPa	119 MPa
Compressive deformation at 27.6 MPa and 50°C, 24 hrs	$\leq 1\%$	0.26%
Ultraviolet transmittance (for 12.5mm of thickness)	$\leq 5\%$	0.16%
Visual Clarity	Must pass readability test	Clear and readable

Total residual monomer		
a) Methyl methacrylate	$\leq 1.6\%$	0.91%
b) Ethyl acrylate		

3.3 O-Ring Material

The O-ring provides a static seal between the viewport and the hull, making it a vital aspect of the sub design. The material of choice has to provide effective under external hydrostatic pressure of 3.02 MPa with appropriate squeeze and groove fill as per ISO 3601. The material must also have good corrosion resistance against seawater and assembly lubricants so that it ages well. The material must be of a homogeneous composition and free from impurities. It must have uniform hardness and maintain dimensional stability and elastic recovery after deformation [10].

After thorough analysis of possible material candidates, the chosen material was NBR (Nitrile Butadiene Rubber). This is a synthetic rubber that was developed during the 1930s. It is fabricated through the co-polymerization of acrylonitrile (ACN) (polar building block) and 1,3-butadiene (nonpolar building block). Its attributes mainly depend on its ACN content. Nitrile rubber exhibits superior mechanical strength with high elasticity as well as good deformation behaviour in a temperature range between -30 °C and 100 °C. Its relatively low heat resistance can be traced to the presence of reactive double bonds. It is used for oil- and fuel-resistant seals such as Simmerrings, O-rings and frame seals. The use of NBR marked the arrival of elastomers as a sealing material particularly in general industry. It is also employed in the auto industry [11].

The following data for our chosen O-ring material (NBR NB702822) was found through Freudenberg's NBR datasheet [12]. This data was analysed alongside what was given in the assignment brief and the properties were used to create a new material in SolidWorks for use in simulation.

Table 3: Physical properties of NBR rubber

Physical Property	ISO 3601-3 Specified Values	Freudenberg NBR NB702822
Hardness (Shore A)	70 ± 5	70 ± 5
Density	Not specified	1170 kg/m ³
Tensile Strength	≥ 12 MPa	18.5 MPa
Poisson's Ratio	Not specified	0.48 - 0.50

3.4 Buoyancy

The volume and mass of the submersible is dependent on the wall thickness.

While the shape of the vessel looks like a sphere, it is not a perfect one. It includes grooves to incorporate the seals. The titanium portion of the hull has a diameter of 1.76m and the acrylic has a diameter of 1.8m. Similarly, we must incorporate the volume of the acrylic and titanium which can be obtained by SolidWorks. This means that the total internal volume of the hull is:

- $V_{total} = \frac{2}{3}\pi r_{titanium}^3 + \frac{2}{3}\pi r_{acrylic}^3 + V_{materials} = 1.427 + 1.523 + 1.4 = 4.35 \text{ m}^3$
- $m_{submersible} = 3965.5 \text{ kg}$
- $F_{submersible} = m_{submersible} \cdot g = 3965.5 \cdot 9.81 = 38,901.56 \text{ N}$
- $F_{water} = \rho_{water} \cdot g \cdot V_{submersible} = 1025 \cdot 9.81 \cdot 4.35 = 43,740.34 \text{ N}$

$\therefore F_{\text{submersible}} < F_{\text{water}}$, as a result, it can be observed that for safety purposes our submersible will have positive buoyancy. The mass properties were obtained from SolidWorks and the value of the mass of the air can be neglected due to its negligible size. For the purpose of this report, the ballast tanks were omitted due to their complexity but in real life application would be used to increase the mass of the submersible to attain negative buoyancy.

$$\frac{F_{\text{sub}}}{F_{\text{water}}} = \frac{38,901.56}{43,740.34} = 0.889 < 3$$

Therefore, as the buoyancy ratio is below 3 it is not too unrealistically buoyant or negatively buoyant. This means that in the event of a system failure the submersible can float back up to safety, and that it is not so low that it would require a large ballast to dive

4. Structural Analysis

4.1 Mesh Refinement

Initially a static analysis study was run for the assembly of the hull and window. To determine if the simulation results were independent of the properties of the mesh, the simulation was run at progressively finer meshes with element sizes halving with each subsequent simulation. The coarsest mesh had 595 elements with a maximum global element size of 360mm. The finest mesh had 419,806 elements with a maximum global element size of 22.5mm. Figure 1 displays the maximum displacement and stress of the model versus the number of elements in the mesh. The maximum displacement of the model shown in Figure 4, converged to 3.49mm indicating that the results for the displacement simulation are independent of the mesh. As the mesh is refined the results for the stress simulation change significantly strongly suggesting the solution for stress depended on the mesh.

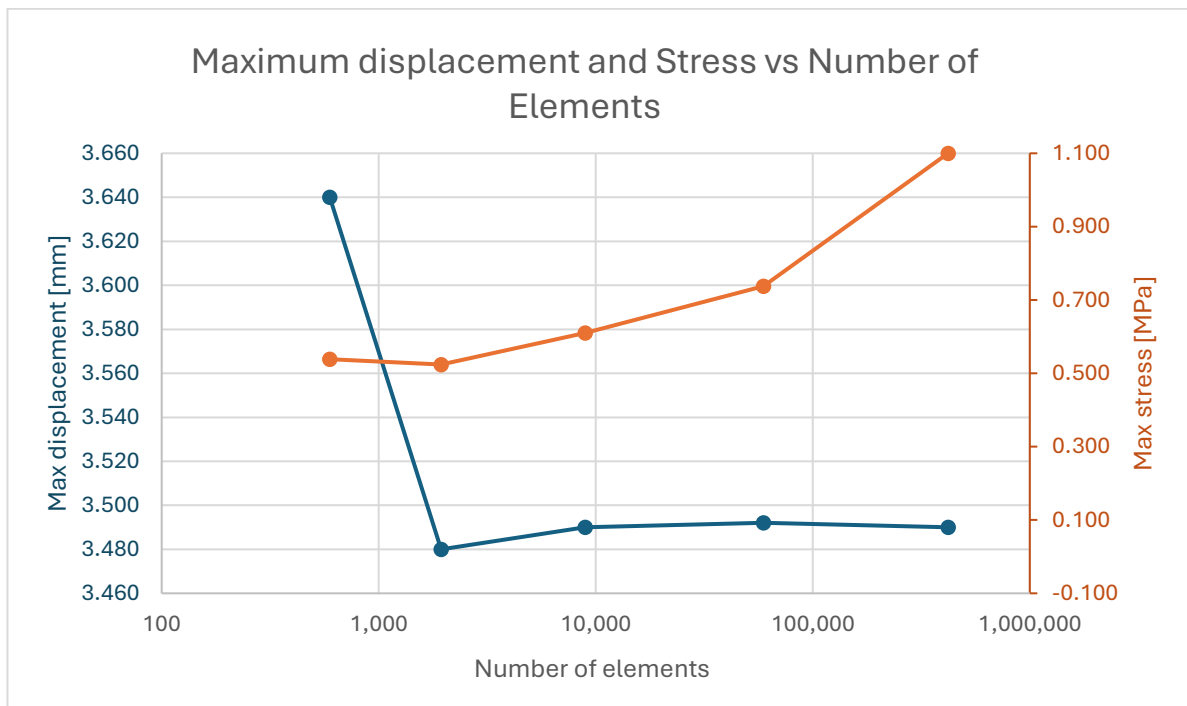


Figure 22: Graph overlaying the plots of maximum stress and displacements against number of elements in the mesh

4.2 Static Analysis

A static linear analysis study of the assembly was used to check for potential failure. See appendix A where Figure 10 displays the results for the maximum von Mises stress in the model with a maximum global element size of 45mm. An external pressure of 3.02MPa was applied to the model and it was fixed with a symmetry fixture and a reference geometry fixture along a radial edge inside the chamber that restricted the model's movement in the X and Y plane. A large stress localised along the reference geometry fixture and increased as the maximum element size decreased. To

mitigate this unrealistic localisation of stress, the window and chamber were evaluated independently of each other. Figure 3 and Figure 4 display the individual simulation results for maximum stress in the window and the chamber. The models were fixed with two symmetry fixtures and a roller slider fixture. A fillet was used to properly capture the stress concentration around the window's flange. These simulations were run with three progressively finer meshes. The maximum von Mises stress in the chamber converged to 31.6MPa which is under the titanium's yield strength of 810MPa. The maximum von Mises stress in the window is converged to 23.8MPa which is under the acrylic's yield strength of 110MPa. These results indicate that the solutions for maximum stresses in the window and chamber are no longer dependant on the mesh as the solution was for the entire assembly.

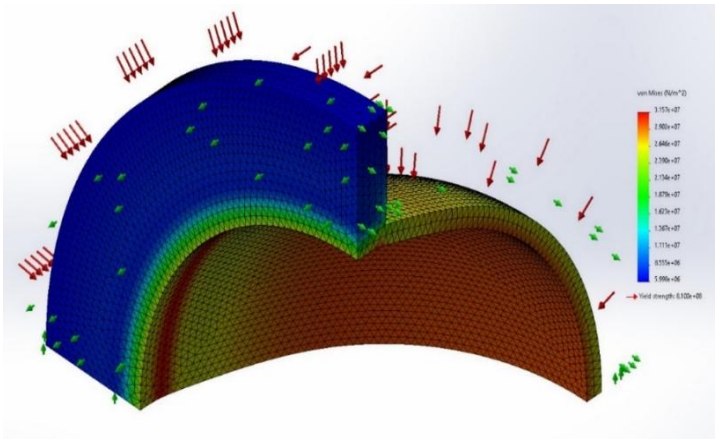


Figure 33: Stress analysis of 1/4 of the chamber

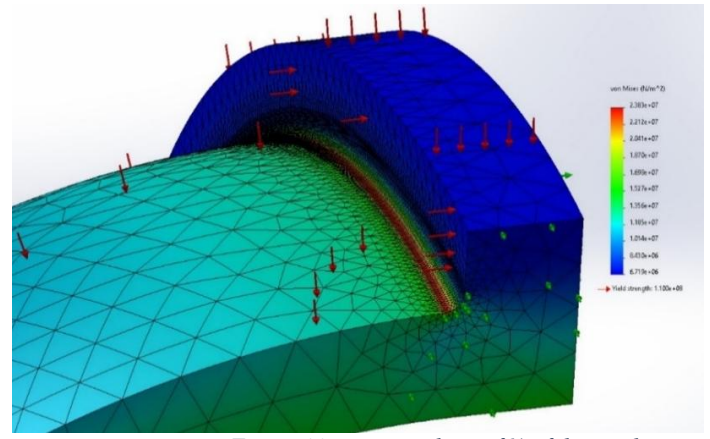


Figure 44: stress analysis of 1/4 of the window

Results for maximum deformation in the static analysis study are shown in Figure 5. The chamber had negligible deformation under the 3.02 MPa external pressure load. The window had a maximum deformation of 3.49 mm which in line with PVHO-1 - 2-2.9.4 (a) requirements for large openings which states radial deformation of the window at maximum external pressure must be less than 2% of the internal diameter.

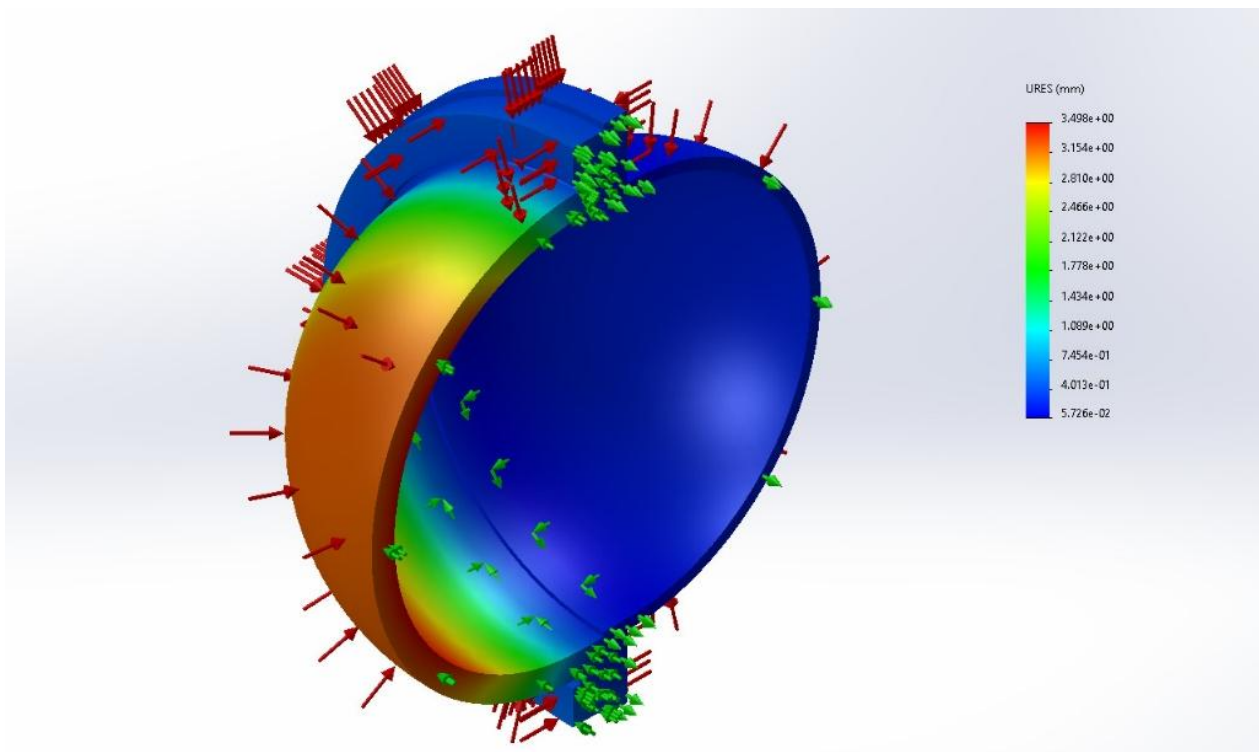


Figure 55: Deformation analysis of the submersible assembly

4.3 Buckling Analysis

For a thin-walled pressure vessel, buckling may happen before yielding. A buckling simulation was run for four modes of deformation. The model was constrained using one symmetry fixture and two reference geometry fixtures on radial edges inside the chamber restricting the vessels movement in the X and Y planes. Mode 1 seen in Figure 6 is the most likely way the chamber will buckle and has a buckling factor of safety of 11.13 meaning the structure will not buckle in this mode shape. The remaining three mode's buckling factors of safety are 11.15, 12.76 and 12.87 respectively, seen in Figure 11, Figure 12 and Figure 13 in Appendix A, meaning the pressure vessel wont buckle in any mode shape.

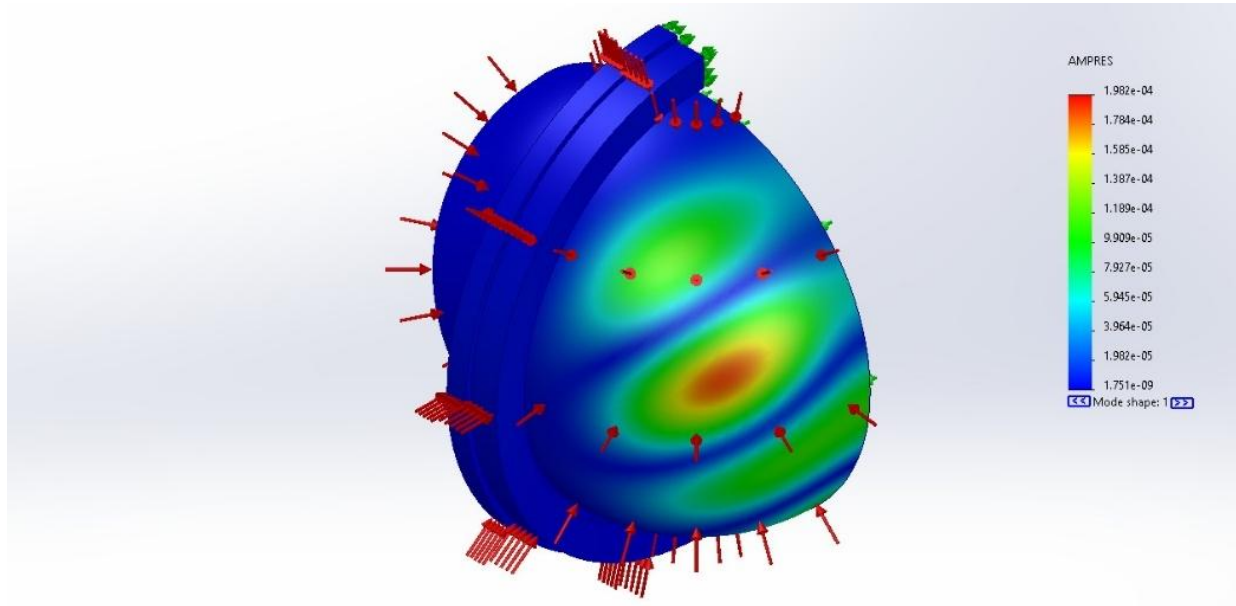


Figure 66: Buckling analysis of the submersible assembly

5. O-Ring Design

The O-ring is part of the sealing system that joins the acrylic hemisphere to the titanium hull using a bolted retaining ring. This creates a watertight joint between the two hemispheres. A static axial seal is used in line with PVHO-1 standards, the O-ring is in a 90-degree triangular crush groove, as shown in figure 7. To minimise stress concentrations and direct metal-to-acrylic contact, gaskets are fitted at the contact points, helping to spread the clamping force evenly across the acrylic surface.

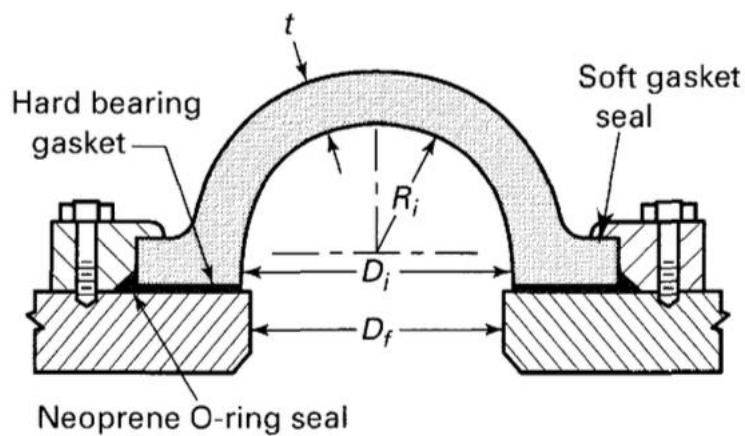


Figure 77: Sectional View of Acrylic Window Assembly Showing O-Ring and Bearing Gaskets

As the vessel is submerged, the external hydrostatic pressure acts on the retaining ring, forcing it against the O-ring in the triangular groove. This causes the O-ring to deform, increasing the contact pressure along both sealing surfaces and

improving the seal as depth increases. The 90-degree groove allows for compression around the circumference. The self-energising effect allows the seal to remain reliable under varying loads and allows for small deflections in the acrylic window without loss of sealing performance [13].

5.1 Model Setup

To model the O-ring, a 0.25-degree circumferential wedge of the full 360 degrees was used. This was used to reduce simulation run time while still accurately modelling the system. The model included the acrylic window, titanium hull, titanium retaining ring and the O-ring positioned in an right angled triangular groove. The triangular groove was chosen as it was advised in the PVHO-1 guide, it offers self-energising behaviour, where the external pressure force the O-ring further into the grove, increasing contact pressure and sealing ability at depth. The soft and hard gaskets weren't modelled as their primary purpose is to distribute the load and prevent metal and acrylic contact. The model is a hyper elastic Mooney-Rivlin material model. The O-ring was defined as a hyper elastic to model its nonlinear deformation characteristics, while the titanium and acrylic were treated as linear elastic materials [14].

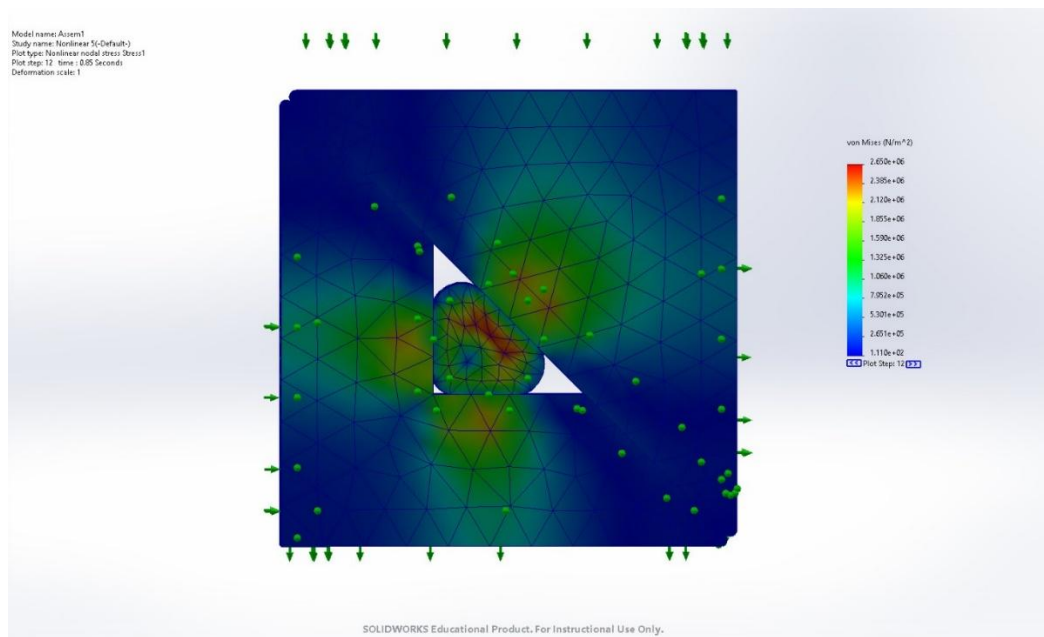


Figure 88: Von Mises Stress Distribution on the O-Ring

Figure 8 shows the von Mises stress distribution on the O-ring, we can see the stress is distributed effectively on both walls, with a concentration on the outer wall where the external pressure acts on the retaining ring. This indicates that as pressure increases the O-ring will be forced deeper into the groove, increasing contact pressure and confirming its self-energising behaviour. The peak von Mises stress was approximately 2.65MPa and indicates that the O-ring stays within the elastic range [12].

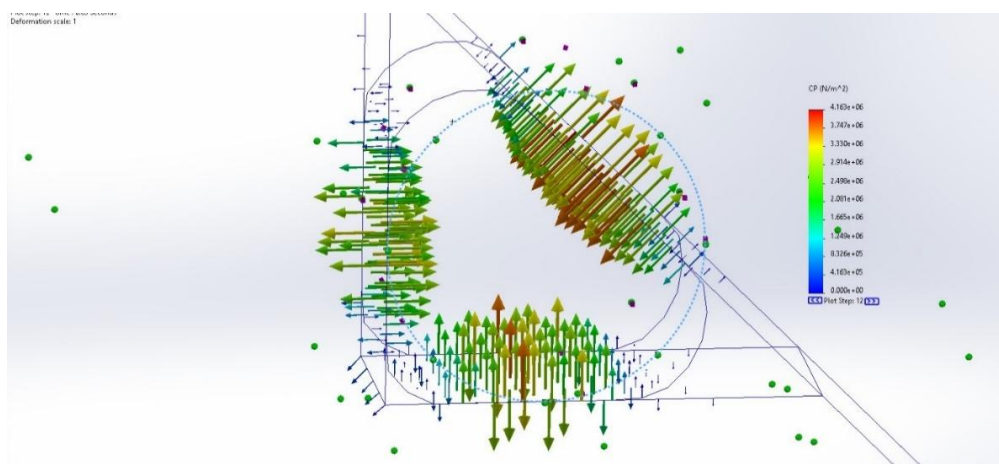


Figure 99: Contact Pressure Distribution and Reaction Vectors on the O-Ring

Figure 9 illustrates contact pressure distribution and reaction vectors acting on the O-ring within the groove. The figure shows that the external hydrostatic pressure compresses the O-ring tightly into the groove, confirming uniform pressure sealing contact and effective pressure transfer.

5.2 Standards

The available O-ring calculators online weren't applicable to the joint we decided to use. To verify if our O-ring was suitable, we took the guidelines from the PVHO-1 [13].

Table 4: Verification of O-Ring Seal Parameters Against PVHO-1 Standards

Verification Parameter	Purpose	Standard	Result
Compression Ratio	“Minimum compression as recommended by the seal manufacturer.”	PVHO-1 - 2-2.11.7	21.4% - pass
Groove Fill	“Groove fill shall not exceed 85 percent to preclude overfilling or extrusion.”	PVHO-1 - 2-2.11.9 (b)	83% - pass
Contact Pressure	“Sufficient surface pressure to maintain seal integrity.”	PVHO-1 - 2-2.11.3	4.1MPa $\geq$ 3MPa - Pass

The parameters above verify that the O-ring satisfies the PVHO -1 sealing requirements. The compression ratio lies within the recommended static axial seal range (20 - 30%)[15]. This ensures sufficient preloading to achieve a seal at sea level, without any self-energisation. The groove fill of 83% complies and prevents any over compression. The contact pressure of 4.1MPa is greater than the external pressure of 3MPa providing a safety factor of 1.37. These results verify that the O-ring will perform without failure at all desired depths.

References

- [1] ‘ISO-21173-2019.pdf’. Accessed: Oct. 12, 2025. [Online]. Available: <https://cdn.standards.iteh.ai/samples/70020/a017e02672584ebcb4ad3c3f94e89bc1/ISO-21173-2019.pdf?utm>
- [2] ‘Design standards for steel spherical pressure hull for a manned submersible | Request PDF’, *ResearchGate*, Aug. 2025, doi: 10.1007/s40722-021-00221-y.
- [3] T. J. Holmquist, ‘Strength and Fracture Characteristics of HY-80, HY-100, and HY-130 Steels Subjected to Various Strains, Strain Rates, Temperatures, and Pressures’, Defense Technical Information Center, Fort Belvoir, VA, Sept. 1987. doi: 10.21236/ADA233061.
- [4] ‘Aluminaut: The World’s First Aluminum Submarine Its Legacy’, Nubeo Watches. Accessed: Oct. 12, 2025. [Online]. Available: <https://nubeowatches.com/blogs/the-aquanova-journal/aluminaut-the-worlds-first-aluminum-submarine-and-its-legacy>
- [5] ‘Carbon/Epoxy Composite Materials - Properties - Supplier Data by Goodfellow’, AZoM. Accessed: Oct. 12, 2025. [Online]. Available: <https://www.azom.com/article.aspx?ArticleID=1995>
- [6] H. Zhang *et al.*, ‘Degradation of Carbon Fiber-Reinforced Polymer Composites in Salt Water and Rapid Evaluation by Electrochemical Impedance Spectroscopy’, *Materials*, vol. 16, no. 4, p. 1676, Feb. 2023, doi: 10.3390/ma16041676.
- [7] ‘Grade 23 Ti 6Al 4V ELI Alloy (UNS R56401)’, AZoM. Accessed: Oct. 12, 2025. [Online]. Available: <https://www.azom.com/article.aspx?ArticleID=9365>
- [8] ‘Titanium Alloy | Titanium 6Al-4V | Ti 6Al-4V | Titanium 64 | Ti 64 | Titanium Grade 5 | Ti Grade 5 | UNS R56400’, Titanium Engineers. Accessed: Oct. 12, 2025. [Online]. Available: <https://www.titaniumengineers.com/ti-6ai-4v.html>
- [9] ‘Transpalite SS Acrylic Specifications’, AcrylicViewPorts.com. Accessed: Oct. 12, 2025. [Online]. Available: <https://www.acrylicviewports.com/transpalite-ss-acrylic-specifications/>
- [10] ‘ISO-3601-1-2012.pdf’. Accessed: Oct. 12, 2025. [Online]. Available: <https://cdn.standards.iteh.ai/samples/58043/f1be17df93fc436ca4e10e4cc09bf030/ISO-3601-1-2012.pdf>
- [11] ‘NBR Meaning - Nitrile Butadiene Rubber’. Accessed: Oct. 12, 2025. [Online]. Available: <https://www.fst.com/materials/nbr/>
- [12] ‘nbr nb702822-en_3.pdf’. Accessed: Oct. 12, 2025. [Online]. Available: https://www.fst.com/-/media/files/material-data-sheets/nbr%20nb702822-en_3.pdf

[13] 'ASME PHVO-1 (2007). Safety Standard for Pressure Vessels for Human Occupancy (1)'.

[14] '4B7 SOLIDWORKS tutorial'.

[15] 'Parker O-Ring Handbook ORD 5700' Available: <https://www.parker.com/content/dam/Parker-com/Literature/O-Ring-Division-Literature/ORD-5700.pdf>

Appendix A: Structural Analysis

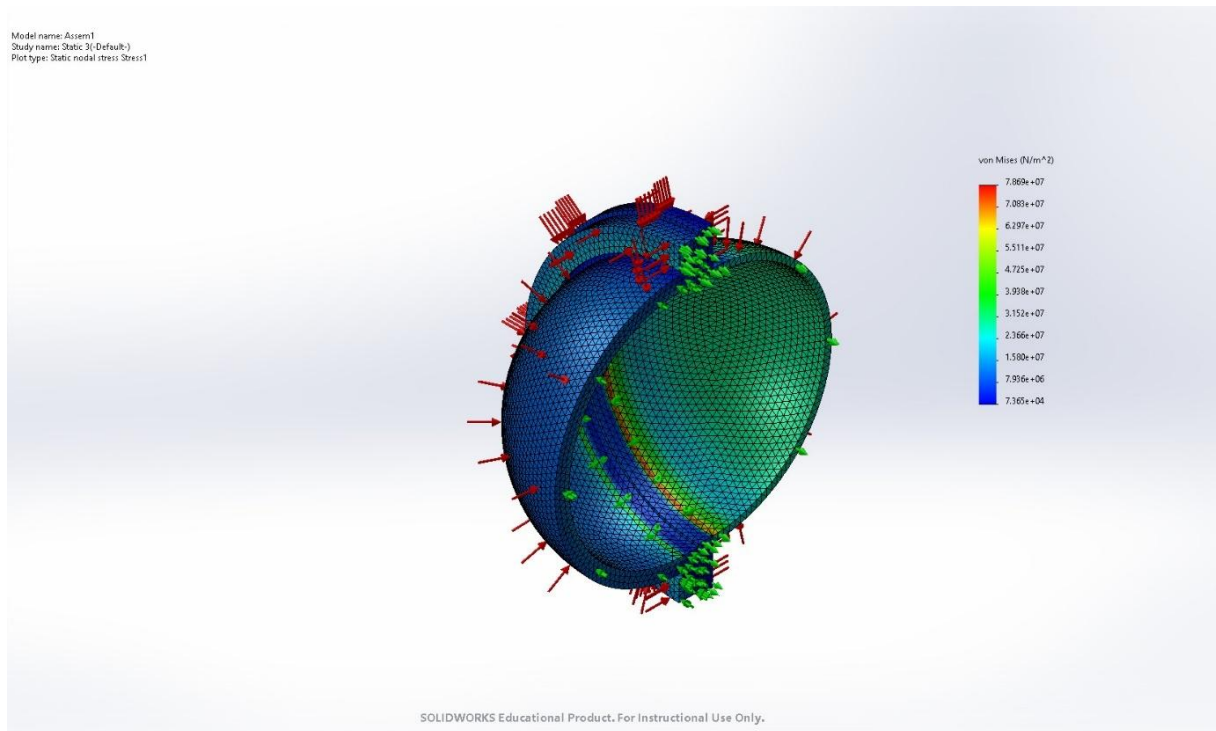


Figure 1010: Stress analysis of submersible assembly

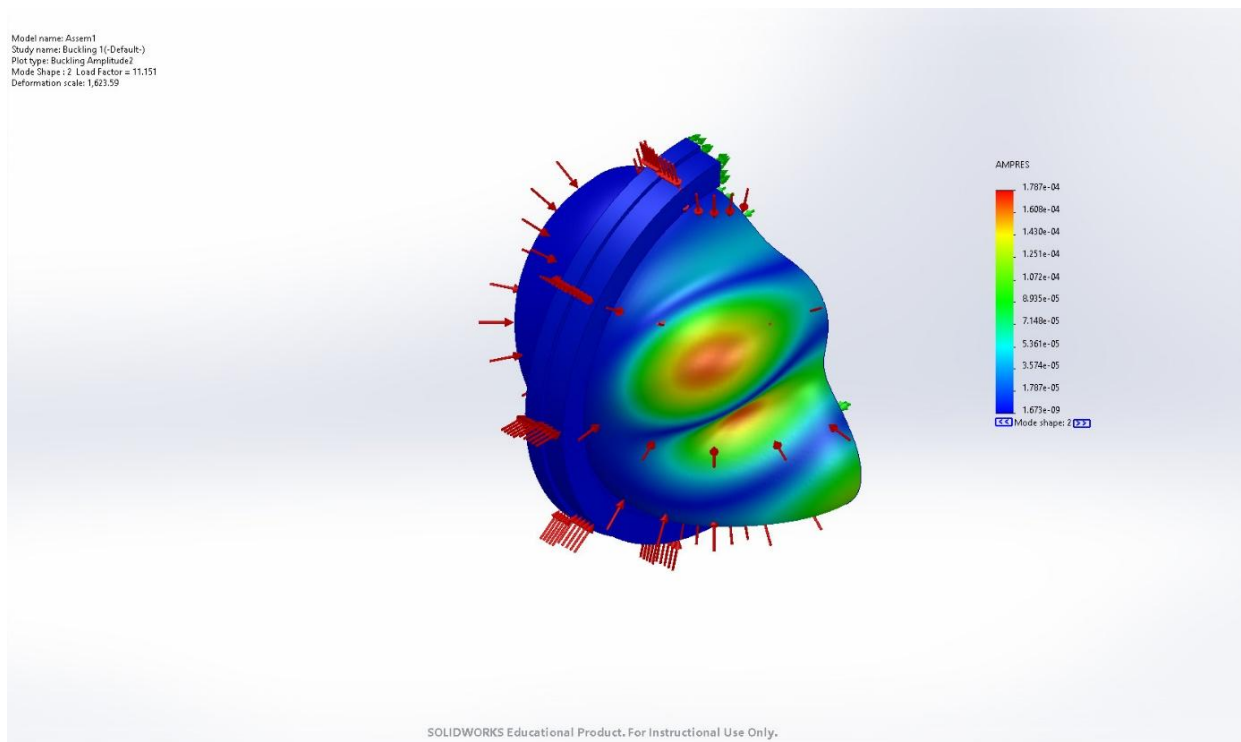
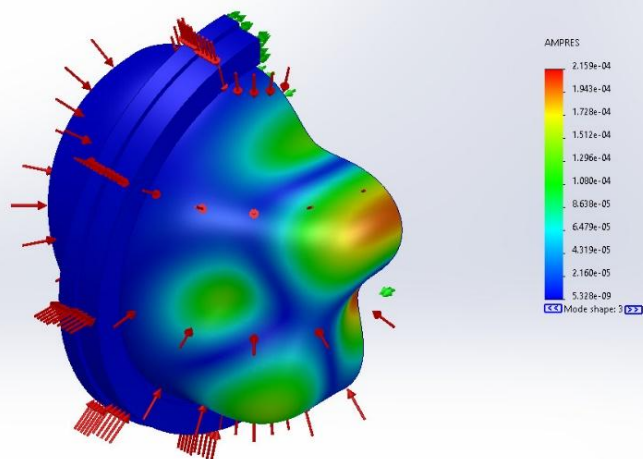


Figure 1111: Mode 2 of buckling analysis of submersible assembly

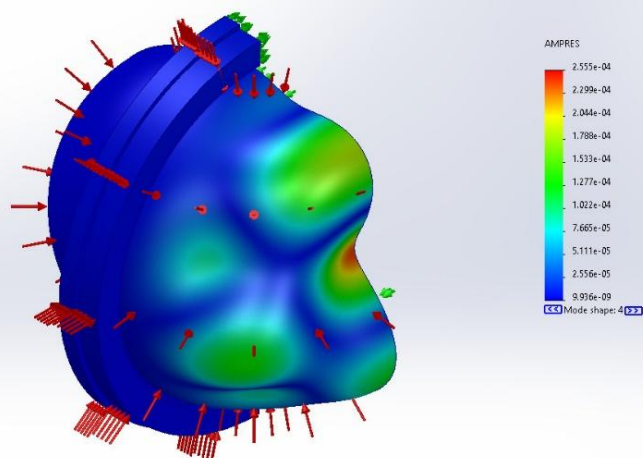
Model name: Assem1
 Study name: Buckling 1 (-Default-)
 Plot type: Buckling Amplitude3
 Mode Shape: 3 Load Factor = 12.762
 Deformation scale: 1,347.41



SOLIDWORKS Educational Product. For Instructional Use Only.

Figure 1212: Mode 3 of buckling analysis of submersible assembly

Model name: Assem1
 Study name: Buckling 1 (-Default-)
 Plot type: Buckling Amplitude4
 Mode Shape: 4 Load Factor = 12.875
 Deformation scale: 976.178



SOLIDWORKS Educational Product. For Instructional Use Only.

Figure 1313: Mode 3 of buckling analysis of submersible assembly

Appendix B: O-ring Design

Compression ratio

$$\text{Compression ratio} = \frac{d_0 - d}{d_0} = \frac{8 - 6.23}{8} = 0.214$$

d_0 = original diameter

d = diameter under compression

The diameter under compression was approximated by taking the average of the diameter at different points.

Housing Fill

$$\text{Housing Fill } (f) = \frac{V_o}{V_g} = \frac{346320}{415562.4} = 0.83$$

V_o = O ring volume

V_g = Volume of the groove

Contact Pressure vs External Pressure.

$$P_c \geq P_{ext} = 4.1 \text{ MPa} \geq 3 \text{ MPa}$$

P_c = Average contact pressure on the sealing faces

P_{ext} = hydrostatic pressure

External Hydrostatic Pressure

$$P_{ext} = \rho_{\text{water}} g h = 1025(9.81)(300) = 3016575 \text{ Pa} \approx 3 \text{ MPa}$$

ρ_{water} = Density of seawater

g = Gravity

h = Depth

Credit Statement

Robert Dillon 22336720: CAD Design, Simulations, Structure Analysis

Magnus Runham 22336288: O Ring Section

Finn O'Connor 22336740: Viewport Material Analysis, Buoyancy, Objectives

Micheal Sadlier 22336873: Materials in the main chamber (titanium), Chamber Geometry, Buoyancy

Peter Hamilton 22337384: CAD Design, material research